ENSO-related climate stress in Chilean copper mining

A mine-cell identification of acute flood and heat shocks and slow water stress, 1960–2025

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Abstract

This report asks when ENSO-related climate anomalies become operationally binding for Chilean copper mines. Each operational channel (FLOOD, HEAT, DROUGHT) is read at two layers: a **General** shock (any-cause threshold crossing, regardless of climatic origin) and an **ENSO-attributable** shock (the subset that also passes a four-condition AND-gate of active ONI, teleconnection-aligned direction, realised local anomaly, and threshold crossing). The contrast between the two layers is the substantive ENSO-attribution analysis. Three findings emerge. *(1) FLOOD:* a wet regional month reduces monthly mine output by approximately 4.7% at the headline regional-mean monthly resolution (significant at the 5% level under wild-cluster bootstrap); the ENSO-attributable subset accounts for roughly half that effect (-2.5%, directionally consistent but not statistically resolved). A conservative mine-cell lower bound shows the same direction at -1.8%. *(2) HEAT:* a robust output null at the mine cell across both layers, consistent with operations adapted to routine heat at the four Antofagasta-basin mines and limited exposure elsewhere. *(3) DROUGHT:* annual output losses are concentrated in *persistent SPEI-12 drought exposure aligned with a matched ENSO-aligned teleconnection gate* ($\hat{\beta} = -0.277$, $p = 0.025$ on the mine-year panel); threshold-only SPEI-12 drought is *not* associated with a mean output decline ($+0.075$, $p = 0.51$), and the streamflow index SSI-12 does not corroborate it (reported as appendix triangulation). The effect survives matched-window teleconnections, gate-definition sensitivity, the fixed-effects ladder, and a lead placebo, but is identified from the post-2020 late megadrought, so it is read as ENSO-aligned persistent megadrought stress rather than a sample-wide drought law. With only seven regional clusters, all acute results are interpreted through wild-cluster bootstrap, leave-one-region, leave-one-event, and ENSO-attribution-rule robustness checks.

1 Question and scope

How much does an ENSO-related climate anomaly that crossed a documented operational threshold for copper mining lower physical copper output in the affected Chilean region, in the month or year it occurred?

1.1 ENSO and the Oceanic Niño Index

ENSO (El Niño–Southern Oscillation) is the coupled ocean–atmosphere oscillation in the equatorial Pacific that shifts the warm-pool position, modulates the Walker circulation, and propagates regional climate anomalies along teleconnection pathways. NOAA's Climate Prediction Center monitors its intensity through the **Oceanic Niño Index** (ONI), a three-month running mean of sea-surface-temperature anomalies in the Niño 3.4 region of the equatorial Pacific.

Standard NOAA thresholds classify a month as **El Niño** when the ONI is at or above $+0.5\,^{\circ}$C, **La Niña** when it is at or below $-0.5\,^{\circ}$C, and **Neutral** otherwise. We adopt those thresholds verbatim throughout the report. The ONI is the first of three named ENSO components that build the stress signal in §2; the other two — the standardised anomaly Z and the regional teleconnection ρ — are introduced there.

1.2 Chile's copper mining geography

The geographic coverage is the seven Chilean regions hosting active copper operations — Tarapacá, Antofagasta, Atacama, Coquimbo, Valparaíso, O'Higgins, and the Santiago Metropolitan Region. Nineteen operations from the Cochilco monthly archive account for roughly 95% of national output. The climate side runs 1960–2025; the regression sample is 4,393 active mine-months over 2006–2025 for acute channels and 369 mine-years for the slow drought channel.

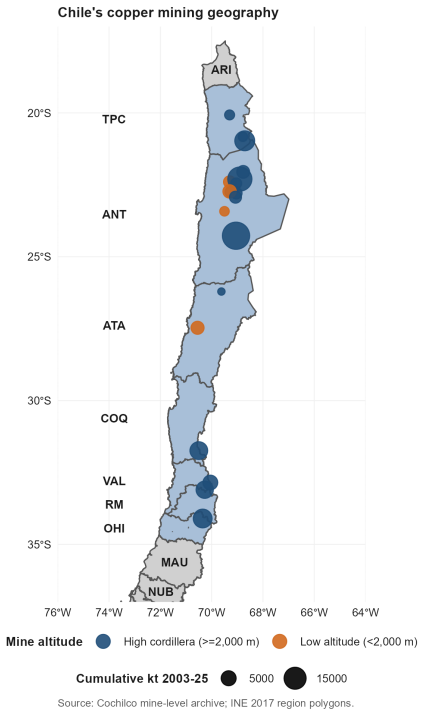


Figure 1: Chile's copper mining geography. Mine sizes are proportional to cumulative output 2003–2025; colours separate the high-cordillera mines (blue, $\geq 2{,}000$ m) from the low-altitude pits (orange, $< 2{,}000$ m). Region codes are the standard three-letter Chilean abbreviations (ARI, TPC, ANT, ATA, COQ, VAL, RM, OHI, MAU, NUB, BIO, ARA, LRI, LLA, AYS, MAG).

Table 1: The 19 copper-mining operations and their key attributes. The final column is the standardised-streamflow (SSI-12) source-basin gauge assigned to each mine; operations drawing from closed endorheic salar systems with no representative streamflow gauge enter the slow water-stress analysis through the mine-cell SPEI-12 only.

Operation	Region	Altitude tier	Elev. (m)	Cum. kt	SSI-12 source-basin gauge
Escondida	Antofagasta	high	3100	26595	Endorheic basin — SPEI-12 only
Codelco - Chuqui, RT y MH	Antofagasta	high	2800	19276	Río Loa (Lequena)
Collahuasi	Tarapaca	high	4400	11384	Río Piga (Collacagua)
Codelco - Teniente	O'Higgins	high	2800	9650	Río Cachapoal (Cortaderal)
Los Pelambres	Coquimbo	high	3600	8237	Río Choapa (Cuncumén)
Anglo American Sur	Santiago Metropolitan	high	3500	7596	Río Maipo (El Manzano)
Codelco - Andina	Valparaiso	high	3800	4845	Río Aconcagua (Chacabuquito)
Spence	Antofagasta	low	1700	3607	Río Loa (Lequena)
Candelaria	Atacama	low	600	3351	Desalinated (seawater) — SPEI-12 only
El Abra	Antofagasta	high	4000	3152	Endorheic basin — SPEI-12 only
Anglo American Norte	Antofagasta	low	1100	2756	Coastal / cross-region — SPEI-12 only
Zaldivar	Antofagasta	high	3000	2661	Endorheic basin — SPEI-12 only
Esperanza	Antofagasta	high	2300	2436	Desalinated (seawater) — SPEI-12 only
El Tesoro	Antofagasta	high	2300	2056	Desalinated (seawater) — SPEI-12 only
Codelco - Gaby	Antofagasta	high	2600	2026	Endorheic basin — SPEI-12 only
Cerro Colorado	Tarapaca	high	2600	1729	Río Huatacondo (Copaquire)
Lomas Bayas	Antofagasta	low	1500	1604	Río Loa (Lequena)
Quebrada Blanca	Tarapaca	high	4400	1567	Desalinated (seawater) — SPEI-12 only
Codelco - Salvador	Atacama	high	2600	1298	Río La Ola (Pedernales)

2 ENSO-related operational stress

This chapter builds the ENSO-related operational stress signal from primitives. §2.1 defines the standardised anomaly Z that measures how unusual a given region-month's climate is against its 1980–2010 baseline; §2.2 introduces the regional Pearson teleconnection ρ that maps the global ONI signal of §1.1 into a regional direction; §2.3 anchors the operational thresholds to mining-engineering literature; and §2.4 combines the three named ENSO components — ONI, Z, ρ — with the operational threshold into a single mine-month stress flag.

2.1 The standardised anomaly Z

A given month's regional climate is expressed as a standardised anomaly,

$$Z^{v}_{r,m,t} \;=\; \frac{X^{v}_{r,m,t} - \mu^{v}_{r,m}}{\sigma^{v}_{r,m}},$$

where $X^{v}_{r,m,t}$ is the observed regional value of variable $v \in \{\text{precipitation},\ \text{mean temperature},\ \text{Tmax}\}$ in region r, calendar month m, year t; $\mu^{v}_{r,m}$ is its 1980–2010 baseline mean; and $\sigma^{v}_{r,m}$ is the 1980–2010 baseline standard deviation. The numerator $X - \mu$ is the raw anomaly in mm or °C; dividing by σ standardises it so that $|Z|$ is comparable across regions and variables with different natural variability. $|Z| = 1$ means the month was one standard deviation away from its 1980–2010 normal; $|Z| = 2$ means two.

The Z-anomaly absorbs *all* sources of variability — ENSO, megadrought, secular warming — and not just El Niño / La Niña events. The ENSO-attribution step in §2.4 isolates the ENSO-related subset by multiplying the Z indicator with the ONI and teleconnection indicators.

2.2 Regional teleconnections ρ

The Pearson teleconnection $\rho^{v}_{r,m}$ is the correlation between the regional climate variable v and the ONI in calendar month m, computed over the 1980–2010 baseline. It is dimensionless, bounded in $[-1, +1]$,

and indicates the *direction* the regional climate is expected to move when the ONI rises (positive ρ: wetter or hotter; negative ρ: drier or cooler). Each region carries twelve monthly Pearson values per variable rather than one seasonal scalar, finer-grained than the seasonal value and consistent with the monthly resolution of the rest of the module.

The 1980–2010 baseline includes pre-megadrought wet decades, so the teleconnection is computed against a climatology that gives all parts of the seasonal cycle a fair share of variability. The Chilean teleconnection literature (Garreaud, Montecinos, Vuille and others) documents that ENSO's regional fingerprint is asymmetric — central-Chile drying under La Niña is steeper than the wetting under El Niño; the symmetric Pearson ρ used here is a defensible first-order proxy, and an asymmetric phase-conditional alternative is reported in the robustness annex. Magnitudes ($|\text{ONI}|$, $|Z|$, $|\rho|$) remain in the master panel as primitive columns for figures, but they are not combined into a composite severity score.

2.3 Operational thresholds

Three channels, all sourced from mining engineering and climate-impact literature: FLOOD has two spatial variants (regional-mean monthly, the headline; mine-cell daily, a conservative point-level lower bound), HEAT has one (mine-cell monthly-mean Tmax), and DROUGHT rests on the meteorological SPEI-12 (a hydrological SSI-12 cross-check is reported in Annex B).

Table 2: Operational thresholds and their roles. The regional-mean monthly accumulation is the headline FLOOD definition; the mine-cell daily variant is a conservative point-level lower bound. Source anchors are listed in the footnote.

Channel	Threshold	Resolution	Role
`FLOOD_regmean_M`[1]	>= 20 mm/month	monthly	FLOOD headline
`FLOOD_mine`[2]	>= 20 mm/day	daily	Conservative point-level lower bound
`HEAT_mine`[3]	>= 23.5 degC monthly mean	monthly aggregate	Acute (robust null)
`DROUGHT_meteo`[4]	SPEI-12 <= -1	monthly aggregated to annual share	DROUGHT headline (meteorological)

[1] Wilcox et al. 2016 GRL 20 mm threshold applied at the regional-mean monthly scale. [2] Wilcox et al. 2016 GRL aluvion definition. [3] Chile DS 594/1999 occupational heat-stress (daily 27 C anchor calibrated to monthly mean on CR2MET 1960-2021). [4] WMO moderate-drought standard (Vicente-Serrano 2010).

FLOOD — regional-mean monthly (headline). The headline FLOOD threshold fires when the area-weighted regional precipitation accumulation in the calendar month reaches ≥ 20 mm. This requires the rain to be *widespread* across the region rather than point-extreme at any single cell, which maps cleanly onto the regional output panel: monthly mine output is itself an aggregate of operations spread across each region's basin and logistics infrastructure, so a "wet regional month" is the right-scaled treatment.

FLOOD — mine-cell daily (conservative lower bound). The mine-cell variant fires when the mine's own 0.05° grid cell receives ≥ 20 mm on any day in the month. This is restrictive because cordillera mines in the Atacama desert and the high altiplano have extreme dry climatologies and rarely receive that much rain in a day; the variant is reported as a conservative lower bound on point-level exposure.

Methodological note — mine-buffered corridor (deferred). A natural intermediate scale — a buffered corridor around each mine's slurry pipeline, access road, and basin — would address the spatial reach of operational disruption (roads, pipelines, tailings dams, valley logistics) more precisely than either the regional mean or the mine cell. Constructing such a mine-specific corridor is a substantial and promising idea that requires dedicated geospatial-attribution work and a separate methodology (mine-buffered

cell masks, distance-decay weighting, or basin-restricted polygons). We reserve it for a future revision.

Channel-matched temporal resolution. The temporal resolution of the threshold and of the standardised anomaly Z used in the ENSO-attribution step are chosen to match each channel's operational mechanism. For FLOOD at the regional-mean monthly scale, both threshold and Z are evaluated at the monthly aggregate. For FLOOD at the daily mine-cell scale, threshold and Z are evaluated at daily resolution (the day the rain falls). For HEAT, the engine uses a monthly aggregation: the threshold is monthly-mean Tmax $\geq 23.5\,^{\circ}$C (calibrated on CR2MET 1960–2021 to the DS 594/1999 daily 27 $^{\circ}$C occupational anchor), paired with the monthly-mean Z. For DROUGHT — a multi-year integrator where what matters is the accumulated water deficit over the year — both the threshold (SPEI-12 ≤ -1, an inherently 12-month index) and the Z are monthly aggregates.

FLOOD validation against documented events. Annex A grades the AND-gate as a binary classifier against 20 documented Chilean flood-disruption episodes 2015–2024, with cohort assignment derived from NOAA ONI and the regional teleconnection direction independently of the AND-gate. Out of 13 events that meet the meteorological definition of an ENSO-related flood and 6 Non-ENSO floods (1 borderline), the FLOOD regional-mean monthly variant catches 38.5% of ENSO events at 100% specificity, and the FLOOD mine-cell variant catches 77% at 100% specificity. The mine-cell ENSO-classifier performance is the stronger of the two; the regional-mean monthly variant's lower sensitivity reflects the resolution-mismatch problem (a 3-day storm in a wet-baseline month averages out at monthly resolution). However, the regional-mean monthly variant produces the more stable *general flood shock* (Spec A) coefficient because it captures the broader wet-month exposure pattern that monthly mine output actually responds to.

The DROUGHT indicator. The slow water-stress channel rests on the WMO standardised meteorological index SPEI-12, computed at the mine cell from local precipitation and potential evapotranspiration: it captures the climatological water balance integrated over twelve months and is defined for every operation. A hydrological streamflow index (SSI-12) was explored as a cross-check but is *not* retained in the main specification — it is physically different from mine-cell meteorological water balance, undefined or inappropriate for many operations (endorheic-salar, desalinated, direct-seawater, and coastal/cross-region), and non-corroborating under the matched gate. It is recorded in Annex B.

2.4 ENSO attribution: three named components

This section combines the three named ENSO components — the ONI (§1.1), the standardised anomaly Z (§2.1), and the regional teleconnection ρ (§2.2) — into a single binary alignment flag, separately for each operational channel $c \in \{\text{DROUGHT}, \text{FLOOD}, \text{HEAT}\}$:

$$\text{ENSOConsistent}^{c}_{r,y,m} = \underbrace{\mathbf{1}\big[\,|\text{ONI}_{y,m}| \geq 0.5\big]}_{\text{(i) active ENSO phase}} \cdot \underbrace{\mathbf{1}\big[\,\text{sign}\big(\rho^{v}_{r,m} \cdot \text{ONI}_{y,m}\big) = s_c\big]}_{\text{(ii) predicted direction}} \cdot \underbrace{\mathbf{1}\big[\,\text{sign}\big(Z^{v,c}_{r,y,m}\big) = s_c\big]}_{\text{(iii) realised Z-anomaly}}. \tag{1}$$

The three indicators multiply to a single binary flag: a region-month is ENSO-aligned only when (i) the ONI sits at or above NOAA's $0.5\,^{\circ}$C active-phase cutoff in absolute value, (ii) the regional teleconnection combined with the ONI's sign predicts the channel-relevant direction s_c, *and* (iii) the realised Z-anomaly actually moves in that direction. The Z-anomaly enters as the third multiplicative indicator; if it disagrees with the predicted direction the flag is zero regardless of how active the ENSO phase is. Channel-specific values of s_c and the Z-anomaly variable used are listed in Table 3.

Table 3: The three multiplied indicators in Equation (1), by channel. DROUGHT and FLOOD share the precipitation Z-anomaly but reverse the sign; HEAT uses the temperature Z-anomaly directly. For FLOOD, the daily-resolution variants (§2.3) use the maximum daily Z on a wet day instead of the monthly-mean Z; for DROUGHT and HEAT, both slow integrators within the month, the monthly-mean Z is used.

Channel	Direction s_c	Z-anomaly variable	Resolution of Z
DROUGHT	-1 (drier-than-normal)	$Z^{\text{pr}}_{r,y,m}$	Monthly mean
FLOOD	$+1$ (wetter-than-normal)	Z^{pr} on wet day	Daily, on day with P $\geq$ 20 mm
HEAT	$+1$ (hotter-than-normal)	$Z^{T}_{r,y,m}$	Monthly mean

The stress flag for channel c in mine-month (i, y, m) is then the AND of the operational threshold (§2.3) and the ENSO-consistency flag in Equation (1):

$$S^{c}_{i,y,m} \;=\; \mathbf{1}\{\text{threshold}^{c}_{i,y,m} \text{ crossed}\} \;\cdot\; \text{ENSOConsistent}^{c}_{r,y,m}.$$

The two factors are constructed independently and the AND step is the only place where they interact. This separation lets us regress mine output on either the broader threshold flag (Spec A in §4) or the narrower ENSO-attributable stress flag (Spec B in §4) cleanly.

3 Data and the master panel

The previous chapter built the stress signal conceptually. This chapter describes the data that feeds it — the mine-level output series and copper price (§3.1), the climate panels at the relevant resolutions (§3.2), and the master panel that joins everything together (§3.3).

3.1 Mine output and copper price

The Cochilco monthly archive reports fine-copper output (kt) for each of the 19 operations from January 2003. Zero-output mine-months are excluded from the regression sample because they reflect either pre-commercial status or whole-mine closure (notably Codelco-Salvador's 2024–2025 Rajo Inca modernisation and Cerro Colorado's 2023–2025 standstill), not month-level climate impact. The resulting active mine-month panel is 4,393 rows. The price control is the LME monthly copper price in USD/mt (logged).

3.2 Climate data and the panel inventory

All climate panels are built from a single source pool: **CR2MET v2.5 1960–2021 plus a 2022–2025 ERA5-Land extension**, with the CR2 Explorer station network at the regional scale. The bias-correction method is panel-specific. The *acute daily* panels (FLOOD, HEAT) are corrected at the mine cell by daily quantile mapping against the 2017–2021 CR2MET daily distribution within each (mine, calendar-month). The *SPEI-12* panel instead uses monthly ERA5-Land 2 m temperature and precipitation, bias-corrected per (mine, calendar-month) on the 2012–2021 overlap (additive for temperature, multiplicative for precipitation; §5.3). Five operational panels are derived from this source pool by applying different spatial aggregations; the inventory is in Table 4.

Table 4: Climate panel inventory.

Panel	Spatial aggregation	Coverage	Feeds
Region monthly accumulation P	Region polygon area-weighted total	1960–2025	FLOOD reg-mean monthly (headline)
Mine-cell daily P, Tmax	Mine 0.05° grid cell	1960–2025	FLOOD mine-cell (conservative); HEAT mine-cell
Mine-cell SPEI-12	Mine 0.05° cell, 12-mo	1960–2021 (CR2) + ERA5 ext. to 2025	DROUGHT (meteorological, headline)
Source-basin SSI-12	Nearest representative gauged catchment	2006–2025	DROUGHT (hydrological; Annex B only)
Region monthly ENSO context	Region polygon, monthly	1960–2025	ONI; ρ; Z; alignment

3.3 The master panel

The result of the data layer is the master panel: one row per mine $\times$ month-start $\times$ channel ($19 \times 792 \times 5 = 75{,}240$ rows), with the columns shown in Table 5. The columns are exactly the primitives introduced in Chapter 2: the physical climate observation and the operational threshold (§2.3); the three named ENSO components — ONI, ρ, Z — whose combination defines the ENSO-alignment flag (§2.4); and the stress flag that is the AND of the threshold and the alignment.

Table 5: Schema of the master panel `ledger/20__master_stress_ledger.csv`. Every stress flag in the report can be re-derived from these columns by inspection.

Column	Content
`mine_operation`, `codregion`, `region_en`, `date`	Mine and region keys; date = first day of the month
`channel`	One of `FLOOD_regmean_M` (headline), `FLOOD_mine` (conservative), `HEAT_mine`, `DROUGHT_meteo`, `DROUGHT_hydro`. Channel roles are defined in `registry/channel_dictionary.csv`.
`physical_value`, `physical_unit`	The observed climate value for the month (mm/day, °C, SPEI unit)
`threshold_value`, `threshold_unit`, `threshold_crossed`	The operational threshold and a 0/1 indicator of crossing
`oni_value`, `oni_phase`	NOAA 3-month rolling ONI and its phase label
`rho_region_month`	Regional teleconnection $\rho^{X}_{r,m}$ between X and ONI in calendar month m
`z_anomaly_value`, `z_anomaly_type`	Realised Z and a label telling whether it is monthly-mean or daily-on-wet-day
`enso_aligned`	0/1 indicator that all three components above point in the channel-relevant direction
`stress_flag`	0/1 product of `threshold_crossed` and `enso_aligned`
`copper_kt`, `log_kt`	Mine output for the month when active

4 Method and identification

4.1 Generalised regression form

For FLOOD and HEAT the regression is at mine-month frequency. The mine-month is indexed by mine i, year y, and calendar-month-of-year m; the region r is determined by i. The general form is:

$$\log(\text{kt})_{i,y,m} \;=\; \alpha_i \;+\; \gamma_{r,m} \;+\; \delta_y \;+\; \beta \cdot X^{c}_{i,y,m} \;+\; \theta \log(\text{price})_{y,m} \;+\; \varepsilon_{i,y,m}.$$

α_i is the mine fixed effect (absorbs each mine's scale, e.g. Escondida $\approx$ 80 kt/month vs Salvador $\approx$ 4 kt/month). $\gamma_{r,m}$ is the region-by-calendar-month seasonal fixed effect (absorbs the region's seasonal output cycle: dry north, wet austral winter in central Chile). δ_y is the year fixed effect (absorbs national or global annual shocks: COVID, copper-price super-cycles, regulation changes). $\theta \log(\text{price})_{y,m}$ controls for within-year monthly LME copper-price fluctuations. Standard errors cluster at the region (7 clusters); inference by CR0 + wild-cluster bootstrap-t.

The focal regressor $X^{c}_{i,y,m}$ takes two forms, run separately:

- **Spec A — General shock** uses the physical threshold flag $\text{Threshold}^{c}_{i,y,m}$ from §2.3. It fires whenever the channel's operational threshold is crossed in mine-month (i, y, m), regardless of cause. $\hat{\beta}^{A}_{c}$ is the average effect of a threshold-firing month on log mine output.
- **Spec B — ENSO-attributable shock** uses the AND-gate stress flag $\text{Stress}^{c}_{i,y,m} = \text{Threshold}^{c}_{i,y,m} \cdot \text{ENSOConsistent}^{c}_{r,y,m}$ from §2.4. It fires only on the subset of threshold-firings that are also ENSO-aligned by the four-condition rule. $\hat{\beta}^{B}_{c}$ is the effect of the ENSO-aligned subset against a comparison group that pools non-ENSO threshold months with genuinely quiet months. This is conservative by construction: if non-ENSO threshold-firing months also reduce output, the pooling biases $|\hat{\beta}^{B}_{c}|$ toward zero.

The contrast between Spec A and Spec B is the central substantive comparison of the report: $\hat{\beta}^{A}$ is the physical effect of the shock regardless of cause; $\hat{\beta}^{B}$ is the share of that effect attributable to the ENSO-aligned subset of events.

4.2 Slow mine-year drought specification

DROUGHT operates on multi-year timescales (aquifer drawdown, reservoir depletion, water-rights tightening). The natural identifying frequency is annual, on a mine-year panel indexed by (i, y):

$$\log(\text{kt}_y)_{i,y} \;=\; \alpha_i \;+\; \delta_y \;+\; \beta \cdot X^{\text{DR}}_{i,y} \;+\; \varepsilon_{i,y},$$

where $X^{\text{DR}}_{i,y} = \overline{S^{\text{DR}}}_{i,y-1:y-3}$ is the three-year cumulative ENSO-aligned SPEI-12 drought-exposure share over years $y-1$, $y-2$, and $y-3$. The annual national copper price is collinear with the year fixed effect δ_y and is therefore dropped from the annual equation (it is retained in the monthly form above).

Caveat on the DROUGHT specification. Unlike the acute channels, the DROUGHT comparison is not a clean Spec A/Spec B share split: the ENSO-aligned regressor is a distinct *persistent-exposure class* (SPEI-12 drought months classified by the matched ENSO-alignment gate of §5.3), reported *against* the all-drought (threshold-only) regressor as a contrast, not as a subset share of a single all-drought coefficient. A hydrological SSI-12 cross-check was explored but is not retained in the main specification (Annex B).

4.3 Inference with seven regional clusters and ENSO-attribution robustness

The panel has only seven regional clusters. Asymptotic cluster-robust standard errors (CR0) underestimate variability with so few clusters; we supplement CR0 with two additional inference steps and an ENSO-attribution-rule robustness battery.

- **Wild-cluster bootstrap-t** (Cameron, Gelbach & Miller 2008). 999 bootstrap draws with Rademacher cluster weights and null-imposed restricted residuals; reported as p_{WCB}. This is the report's small-sample-honest p-value.

- **Leave-one-region-out (LOO).** Re-fit on each subsample of six regions and report the range of the coefficient and its p-value. Identifies whether the result is driven by a single region.
- **Leave-one-event-out.** Re-fit dropping each of three documented multi-year ENSO event windows (2015–16 El Niño, 2020–22 La Niña, 2023–24 El Niño).

ENSO-attribution robustness. The baseline rule (Equation 1) uses the 3-month rolling ONI, a symmetric Pearson teleconnection ρ, and the realised Z. Three alternatives test whether the headline coefficients depend on this construction:

1. **Seasonal-window ONI.** The 3-month rolling ONI moves month-by-month; the Chilean teleconnection literature often summarises ENSO by a channel-relevant seasonal aggregate (JJA for the central-Chile rainy season; DJF for the austral-summer temperature signal). We re-fit replacing the rolling ONI with JJA ONI for DROUGHT and FLOOD and DJF ONI for HEAT.
2. **Spearman ρ.** Pearson correlation is sensitive to outliers; Spearman is rank-based. We re-compute the regional teleconnection by Spearman on the same 1980–2010 baseline and re-fit.
3. **Phase-conditional $\overline{Z}$.** The symmetric sign test $\text{sign}(\rho \cdot \text{ONI}) = s_c$ is replaced by the phase-conditional mean $\overline{Z}^{v}_{r,m,\text{phase}}$ computed on 1980–2010 separately for El Niño and La Niña phases. The predicted direction in a given month is $\text{sign}(\overline{Z}^{v}_{r,m,\text{phase of that month}})$. This captures the asymmetric Chilean fingerprint — central-Chile drying under La Niña is steeper than the wetting under El Niño — that a symmetric Pearson sign-test cannot.

The ENSO-attribution robustness checks confirm that the FLOOD Spec B coefficient is sign-stable across all three alternative rules. The HEAT mine-cell coefficient is a robust null across all four rules. The DROUGHT 3-year-cumulative SPEI-12 coefficient is sign-stable under the Spearman and phase-conditional alternatives but flips sign under the seasonal-window JJA ONI, which discards the within-year variation that a multi-year accumulator needs — a known weakness of channel-specific seasonal aggregates for slow channels.

The robustness re-fits are not perfectly apples-to-apples with the headline Spec B. The headline regresses on the stress flag alone (non-ENSO threshold months pooled into the reference); the robustness script regresses on both a non-ENSO threshold flag and the stress flag, separating the two within the same equation. The reported alternative-rule coefficient is the partial effect of the stress flag in that two-regressor specification, holding the non-ENSO threshold effect fixed. This is a slightly less conservative comparison than the headline; the sign-stability conclusion holds under either partition.

5 Results

5.1 FLOOD: wet regional months reduce mine output; about half the point estimate falls in the ENSO-aligned subset

Table 6: Acute mine-month regressions, reframed. Spec **A — General** fires on any physical threshold crossing; Spec **B — ENSO-attributable** fires only on the AND-gate (active ONI, teleconnection-aligned, realised-Z direction, threshold crossed). Two p-values are reported: CR0 (asymptotic cluster-robust at region, 7 clusters) and WCB (wild-cluster bootstrap-t, Rademacher, null imposed, $B = 999$). LOO range is the minimum and maximum coefficient across the seven leave-one-region refits. Mine-cell Spec B uses the singletons-filtered regressor (the 4 mines that fire only once at the mine cell are dropped because under mine FE a singleton firing has no within-mine identifying variation; the raw coefficient is $+0.014$, $p_{\text{WCB}} = 0.61$).

Specification	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	N	LOO range
FLOOD — regional-mean monthly (headline)						
General flood shocks	-0.047	0.010	0.002***	0.013**	4,393	[-0.058, -0.042]
ENSO-attributable flood shocks	-0.025	0.020	0.246	0.287	4,393	[-0.038, -0.013]
FLOOD — mine-cell (conservative)						
General flood shocks	-0.018	0.016	0.301	0.395	4,393	[-0.023, -0.007]
ENSO-attributable flood shocks (singletons filtered)	-0.002	0.028	0.939	0.978	4,393	[-0.018, 0.024]
HEAT — mine-cell						
General heat shocks	-0.003	0.009	0.713	0.686	4,393	[-0.031, -0.000]
ENSO-attributable heat shocks	-0.018	0.009	0.087*	0.700	4,393	[-0.099, -0.014]

Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Each FLOOD variant is reported under two specifications: **Spec A — General flood shocks** (the any-cause threshold flag, identifying the average effect of a flood-threshold violation on monthly output regardless of cause), and **Spec B — ENSO-attributable flood shocks** (the AND-gate stress flag, identifying the effect of the ENSO-aligned subset against an unrestricted comparison group). The contrast Spec A vs Spec B is the substantive ENSO-attribution analysis.

Headline: FLOOD regional-mean monthly, Spec A. $\hat{\beta} = -0.047$ ($p_{\text{CR0}} = 0.003$, $p_{\text{WCB}} = 0.013$). A wet regional month reduces monthly mine output by approximately 4.7%, significant at the 5% level under wild-cluster bootstrap on 7 clusters. The LOO-region range is $[-0.058, -0.042]$ — entirely negative, no single cluster drives the result. This is the most stable and most identified single FLOOD finding in the module.

ENSO-attributable subset at the regional mean: Spec B. $\hat{\beta} = -0.025$ ($p_{\text{CR0}} = 0.246$, $p_{\text{WCB}} = 0.287$). Roughly half the general-flood effect is concentrated in months that also pass the AND-gate (active ONI, teleconnection-aligned, realised-Z direction); the remaining $\approx -2.2\%$ is the non-ENSO subset. The Spec B coefficient is directionally consistent but not statistically resolved under wild-cluster bootstrap on 7 clusters — the AND-gate fires on only 115 mine-months and the comparison group pools the non-ENSO threshold months with genuinely quiet months, which biases $|\hat{\beta}^B|$ toward zero.

Mine-cell variant: conservative point-level lower bound. Spec A: $\hat{\beta} = -0.018$ ($p_{\text{CR0}} = 0.30$, $p_{\text{WCB}} = 0.40$), directionally consistent with the regional-mean headline but smaller and not statistically resolved — many of the regional wet months do not deposit ≥ 20 mm/day at the high-altitude mine cell itself. Spec B (with singleton-firing mines filtered, see Table 6): $\hat{\beta} = -0.002$ ($p_{\text{CR0}} = 0.94$, $p_{\text{WCB}} = 0.98$) — statistically indistinguishable from zero, as expected when only 65 firings concentrate in 4 mines.

What about an intermediate scale? A mine-buffered corridor (between the broad regional mean and the narrow mine cell) would be the natural intermediate spatial scale — defined by a buffer around each mine's slurry pipeline, access road, and basin. Constructing it requires dedicated geospatial-attribution work and is reserved for a future revision (§2.3). Within the current report, the regional-mean headline and the mine-cell lower bound indicate the likely spatial reach: the true mine-buffered corridor coeffi-

cient plausibly lies somewhere between -4.7% (regional) and -1.8% (point-level), though a corridor that also captures pipeline, access-road, and valley disruption is a physically distinct exposure mechanism and could fall outside that range.

The honest reading: the FLOOD channel has a clear, stable physical signature. Wet regional months reduce monthly mine output by roughly $4-5\%$; roughly half of that effect is concentrated in months that meet the AND-gate's ENSO-attribution criteria. The ENSO subset's reduced statistical resolution reflects the small firing count rather than a substantive disagreement with the physical-shock magnitude.

5.2 HEAT: null at the mine cell

The mine-cell HEAT coefficient is essentially zero under both Spec A (any-cause monthly-mean Tmax $\geq 23.5\,^{\circ}$C: $\hat{\beta} = -0.004$, $p_{\text{CR0}} = 0.71$, $p_{\text{WCB}} = 0.69$) and Spec B (ENSO-attributable: $\hat{\beta} = -0.018$, $p_{\text{CR0}} = 0.09$ but $p_{\text{WCB}} = 0.70$ — the asymptotic significance does not survive the wild-cluster bootstrap). The ENSO-attribution robustness checks confirm the null across all three alternative rules. The firing pattern by mine is bimodal in active mine-months: four operations cross the monthly-mean threshold in essentially every month (Anglo American Norte, Spence, Lomas Bayas, and Codelco-Gaby, all in the hot Antofagasta basin, ≈ 99–100%); Zaldivar crosses it in $\approx 62\%$ of months, Candelaria in $\approx 21\%$, El Tesoro in $\approx 8\%$, and Chuquicamata in $< 1\%$; the remaining eleven operations — including the high-cordillera majors and the Atacama mid-altitude pits — never do. The mines that routinely face heat exposure show no measurable monthly output response to it, consistent with operational practices built around that exposure; monthly resolution cannot test whether finer-frequency productivity losses (worker absenteeism, shift slowdown) are present underneath.

5.3 DROUGHT: ENSO-aligned persistent water stress lowers annual output

Why DROUGHT uses a different clock from FLOOD and HEAT. The three climate channels use the same organizing principle—a physical threshold first, then an ENSO-alignment rule—but they operate on different time scales. FLOOD and HEAT are acute shocks: a high-rainfall or high-temperature event can be classified in the same month in which the threshold is crossed. DROUGHT is a slow stock, not a one-month event. SPEI-12 observed in month m summarizes the previous twelve months of water balance, and mine output responds through accumulated water scarcity, reservoir and groundwater constraints, water-rights pressure, and adaptation decisions. For that reason, the DROUGHT specification uses annual mine output and a three-year lagged cumulative exposure measure. The ENSO rule is also evaluated over the drought accumulation window rather than treated as a purely contemporaneous monthly classifier. This is not a different standard of evidence; it is the same threshold-and-alignment logic matched to a slower physical mechanism.

Data correction (2022–2025). The mine-cell SPEI-12 is constructed from CR2MET for 1960–2021 and extended through 2025 with ERA5-Land monthly 2 m temperature and precipitation at each mine cell, bias-corrected per (mine, calendar-month) against the CR2MET climatology on the 2012–2021 overlap. The historical series is preserved exactly: recomputed 1960–2021 SPEI-12 correlates 1.000 with the original series. The 2022–2024 extension tracks the independent regional SPEI closely ($r \approx 0.91$) and recovers the severe 2023–2024 northern drought. Missing mine-cell SPEI is coded as missing, not as zero drought; annual drought shares require twelve valid months; and the three-year cumulative regressor requires a complete lag window.

The central contrast: not all drought, but ENSO-aligned persistent drought. On the mine-year panel (mine + year FE, standard errors clustered by region), the three-year cumulative share of *all* SPEI-12 drought months is not associated with lower annual output: $\hat{\beta} = +0.075$ ($p = 0.510$), and it is essentially zero once mine-specific trends are absorbed. The damaging exposure is the subset classified by the matched ENSO-aligned gate: months in which SPEI-12 drought coincides with active ENSO conditions, a teleconnection-predicted dry direction, and a realised regional precipitation deficit. Under the final matched-ρ gate, that exposure is negative and statistically resolved ($\hat{\beta} = -0.277$, $p = 0.025$; -0.529, $p = 0.035$ with mine-specific trends). In the two-regressor decomposition, the ENSO-aligned compo-

nent remains negative (-0.293, $p = 0.027$), while the complementary non-ENSO drought component is positive and not significant ($+0.162$, $p = 0.254$). The substantive reading is *not* that "ENSO drought is larger than all drought." It is that the broad pool of SPEI-12 drought months is heterogeneous—including routine, anticipated, or locally adapted dryness—whereas the output penalty is concentrated in persistent drought exposure that is aligned with the ENSO-aligned teleconnection.

Table 7: DROUGHT: the all-drought null versus the ENSO-aligned negative, and the matched-scale robustness battery. "ENSO-aligned" = SPEI-12 drought classified by the matched 12-month ENSO-aligned gate (active ONI, teleconnection-predicted dry direction with matched-window ρ_{12}, realised regional precipitation deficit). Mine-year panel (19 operations, $N = 369$); mine + year FE; SE clustered by region.

Specification	$\hat{\beta}$	p_{CR0}	Reading
Headline contrast (mine + year FE)			
All-drought (threshold-only SPEI-12)	$+0.075$	0.510	no mean output effect
ENSO-aligned, matched-ρ	-0.277	0.025	ENSO-aligned subset (negative, resolved)
ENSO-aligned, + mine trends	-0.529	0.035	larger, but more parameterised (fragile)
Two-regressor decomposition (mine + year FE)			
ENSO-aligned component	-0.293	0.027	negative
non-ENSO component	$+0.162$	0.254	null / positive
Matched-scale diagonal (SPEI-k $\times$ k-month gate)			
SPEI-3 $\times$ 3-month	-0.429	0.037	negative, resolved
SPEI-6 $\times$ 6-month	-0.447	0.061	(multi-month)
SPEI-12 $\times$ 12-month	-0.277	0.025	
Gate-definition sensitivity (SPEI-12)			
drop realised-precipitation condition	-0.266	0.024	barely moves
drop teleconnection direction	-0.101	0.212	halves; teleconn. essential
realised dryness only (no ENSO)	$+0.071$	0.501	null
Period sensitivity and placebo			
pre-2020 output years only	$+0.801$	0.131	no pre-megadrought signal
post-2020 output years only	-0.615	0.051	identification block
lead placebo ($y{+}1..y{+}3$ exposure)	$+0.094$	0.549	future does not predict
Leave-one-region-out and randomisation inference (matched-ρ ENSO-aligned)			
LOO-region range (7 refits)	$[-0.443, -0.211]$		all negative; $p < 0.10$ in 6/7
randomisation-inference p (B=2000)	0.134		one-sided; WCB unstable on N=369

Time-scale matching (construct validity). Because SPEI-12 is a twelve-month accumulated index, the ENSO gate is evaluated over a matched window: a k-month ONI, a k-month precipitation anomaly, and a teleconnection ρ_k recomputed over the same k-month window on the 1980–2010 baseline. This check addresses the concern that a twelve-month drought stock was being classified with a one-month ENSO condition. The matched-ρ battery supports the slow-channel interpretation. The ENSO-aligned coefficient is negative at the multi-month horizons (SPEI-3: -0.429, $p = 0.037$; SPEI-6: -0.447, $p = 0.061$; SPEI-12: -0.277, $p = 0.025$), while all-drought remains null at each scale (Figure 2). The result is also not a single AND-gate artifact: dropping the realised precipitation condition barely moves the SPEI-12 coefficient (-0.266, $p = 0.024$), but dropping the teleconnection direction halves it and makes it insignificant (-0.101, $p = 0.212$), and realised dryness alone is null ($+0.071$, $p = 0.501$). A lead placebo also passes: future ENSO-aligned drought exposure ($y + 1$ to $y + 3$) does not predict current output ($+0.094$, $p = 0.549$), whereas lagged exposure does.

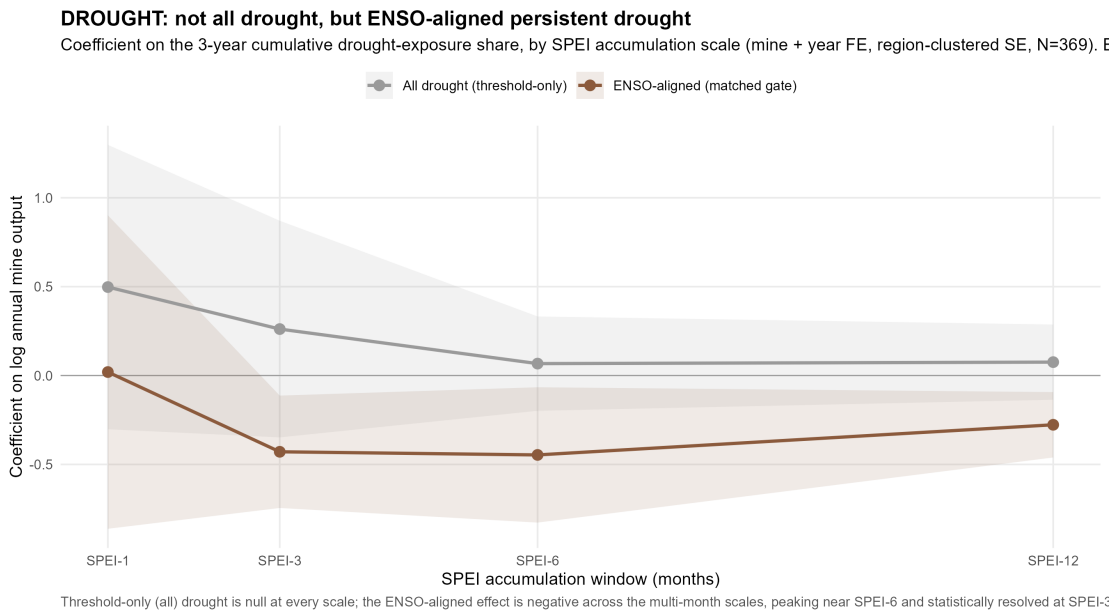


Figure 2: DROUGHT: the all-drought null versus the ENSO-aligned negative, by SPEI accumulation scale. Each point is the mine-year coefficient on the three-year cumulative drought-exposure share (mine + year FE, region-clustered SE, $N = 369$); the ENSO-aligned series uses the matched k-month gate. Threshold-only (all) drought is null at every scale, while the ENSO-aligned effect is negative across the multi-month scales, peaking near SPEI-6 and statistically resolved at SPEI-3 and SPEI-12, consistent with a slow water-stress channel rather than a short-window artifact. Every accumulation window plausibly re-detects the same post-2020 signal, which is why SPEI-3 is not weaker than SPEI-12 despite the slow-integrator framing.

Effect size. To make the coefficient interpretable, note the spread of the regressor. The matched-ρ ENSO-aligned exposure share is concentrated: it is exactly zero in 79% of mine-years (no ENSO-aligned persistent drought in the three-year window), with a mean of 0.05, a standard deviation of 0.12, and a maximum of 0.75; among the 77 exposed mine-years it averages 0.23. Because the distribution is zero-inflated, its interquartile range is degenerate (zero), so the per-standard-deviation and tail moves are the informative summaries. The preferred coefficient ($\hat{\beta} = -0.277$) implies that a $+10$ percentage-point increase in the share is associated with roughly a 2.8% lower annual mine output; a one-standard-deviation ($+0.12$) increase with about a 3.4% decline; and a move from no exposure to the 90th percentile (0.19) with about a 5.4% decline. The most-exposed mine-years (share near 0.75) carry an implied loss on the order of 20%. The specification with mine-specific linear trends yields a larger coefficient ($\hat{\beta} = -0.529$), but this is a more heavily parameterised and more fragile specification — about nineteen mine-specific trends on top of nineteen mine and twenty year fixed effects, on $N = 369$ — and should be read as larger, not as an upgrade in confidence; we keep -0.277 as the headline. These magnitudes should be interpreted as changes in exposure intensity, not as the effect of a single drought month.

What identifies the effect (the binding caution). The negative result is identified from the post-2020 late megadrought. Dropping output years 2022–2025 changes the coefficient to $+0.621$ ($p = 0.182$); dropping 2021–2025 gives $+0.701$ ($p = 0.154$); and the pre-2020 sample gives $+0.801$ ($p = 0.131$). By contrast, the post-2020 window alone gives -0.615 ($p = 0.051$). The broader 37-operation Annuario panel (1988–2024) reproduces the negative sign and similar magnitude under the matched SPEI-12 gate, but does not reach significance on the noisier sample. The DROUGHT result should therefore be framed as an ENSO-aligned persistent megadrought-stress effect, not as a sample-wide law that all drought generally lowers mine output and not as proof that ENSO caused the megadrought.

Leave-one-region-out and randomisation inference. Because the drought channel is the least cleanly identified of the three, we subject it to the same small-sample scrutiny as the acute channels (Table 7,

final block). The leave-one-region-out battery re-fits the matched-ρ ENSO-aligned regression seven times, each dropping one region: the coefficient stays negative in all seven refits, ranging from -0.211 to -0.443, and remains significant at the 10% level in six of the seven (at the 5% level in three). The single refit that loses significance is the one dropping Antofagasta ($\hat{\beta} = -0.44$, $p = 0.45$), but that is because Antofagasta alone supplies roughly half the mine-years (N falls from 369 to 177): the point estimate there is if anything more negative, not smaller, so no single region is manufacturing the result. For a p-value that does not lean on a seven-cluster variance estimate, we use randomisation inference: holding the mine and year fixed effects fixed and permuting the ENSO-aligned exposure across mines within each year, $B = 2000$ times, the share of placebo coefficients at least as negative as the observed -0.277 is 0.13 (one-sided). This assumption-free figure is markedly more conservative than the asymptotic $p = 0.025$ and places the effect near the 10–15% level. We rely on randomisation inference here rather than the wild-cluster bootstrap because the bootstrap does not stabilise on this panel: with only seven clusters and an exposure regressor that is zero in roughly four-fifths of mine-years and concentrated in a single multi-year block, the null-imposed bootstrap distribution is degenerate. The honest summary is that the sign is robust and the magnitude stable across regions, but the significance is borderline once one stops relying on asymptotics — consistent with a real effect identified from a single megadrought rather than a precisely estimated structural parameter.

Streamflow cross-index. A hydrological SSI-12 cross-check was explored but is not retained in the main specification: it measures source-basin streamflow rather than mine-cell meteorological water balance, is unavailable or physically inappropriate for many operations (closed-basin, desalinated, direct-seawater, coastal, and mixed-supply), and does not corroborate the meteorological result under the matched gate. On the small gauged-basin subset the ENSO-aligned streamflow coefficient is in fact *positive* ($\hat{\beta} = +0.501$, $p = 0.065$, $N = 148$) — the opposite sign to the meteorological SPEI-12 result. We do not read this as evidence against the SPEI-12 finding: SSI-12 measures a physically different exposure (source-basin streamflow rather than mine-cell water balance) on a small, non-randomly selected subset of operations, and the megadrought's effect on managed reservoir releases need not track the meteorological water balance at the mine. The disagreement is, however, stated plainly here rather than confined to the annex: the slow channel rests on SPEI-12 alone, with SSI-12 as a non-corroborating cross-check. It is treated as a data limitation and recorded in Annex B.

Box 1 — Case study (mechanism support, not econometric proof): ENSO-aligned SPEI-12 exposure during the late megadrought.

The matched SPEI-12 ENSO-aligned exposure share spikes in 2021–2024 across the central–northern copper regions, while all-drought exposure remains broader and is dominated by non-ENSO drought in many years. This is the period in which Chilean copper faced its most visible water-supply pressure, documented in the COCHILCO *Consumo de Agua en la Minería del Cobre* series: continental (freshwater) withdrawals fell while desalinated and direct-seawater supply rose sharply, and several major operations accelerated seawater transitions. The contrast is consistent with adaptation: routine SPEI-12 drought can be absorbed by alternative water supply (desalination, recirculation) and so leaves no mean output footprint, while persistent, broadly teleconnected ENSO-aligned episodes coincide with the years and operations where water constraints actually bind. This box is mechanism support; the identification rests on the panel estimates in Table 7.

Inference note. Drought p-values are CR0 cluster-robust on seven regions. With so few clusters, inference does not rest on one p-value alone: the evidentiary package is the combination of (i) the all-drought null versus ENSO-aligned negative contrast, (ii) the matched-window ρ_k check, (iii) gate-definition sensitivity, (iv) the lead placebo, and (v) transparent disclosure that identification comes from the late-megadrought block.

5.4 Coefficient summary across channels and inference battery

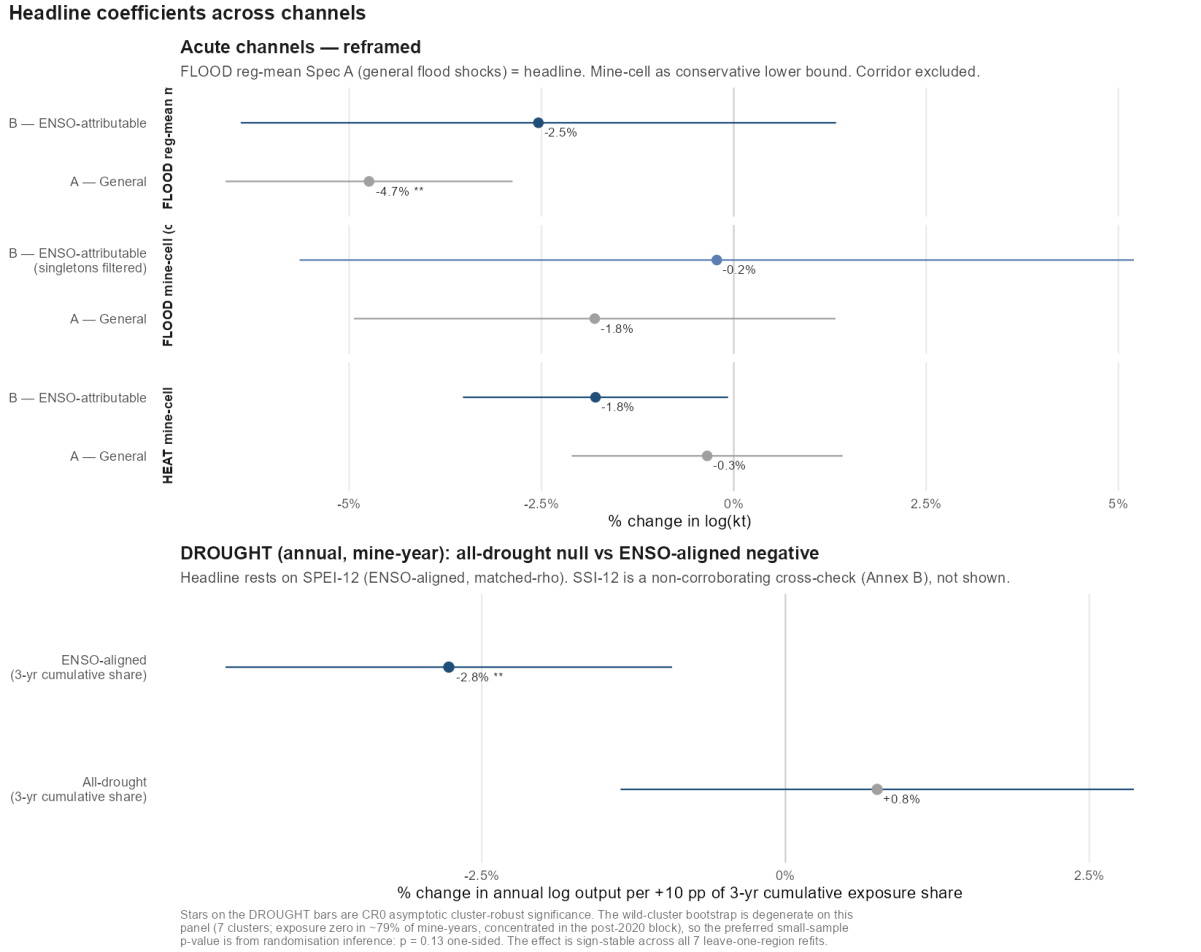


Figure 3: Forest plot of headline coefficients. Acute rows (top): % change in monthly log mine output per 0→1 firing of the flag. Spec A = General shock (any-cause threshold); Spec B = ENSO-attributable shock (AND-gate). Drought rows (bottom): the all-drought (null) and ENSO-aligned coefficients, as % change in annual log mine output per $+10$ percentage-point change in the three-year cumulative drought-exposure share. Bars = 95% asymptotic cluster CI at region. Significance: WCB for acute channels. DROUGHT stars are CR0 asymptotic; the wild-cluster bootstrap is degenerate on the annual panel, so randomisation inference gives $p \approx 0.13$ one-sided (§5.3). Mine-cell Spec B uses the singletons-filtered regressor (4 singleton-firing mines dropped; raw coefficient $+1.4\%$, $p_{\text{WCB}} = 0.61$).

The forest plot makes the General-vs-ENSO-attributable comparison visible. For FLOOD at the regional-mean monthly headline, Spec A returns $-4.7\%^{**}$ and Spec B returns -2.5% in the same direction — roughly half the general-flood effect is concentrated in the AND-gate's ENSO subset. The mine-cell variant is consistent in direction (Spec A: -1.8%) but the firing count at the point level is too thin to identify the ENSO subset separately (-0.2%, indistinguishable from zero after singleton filtering). HEAT is a null at the mine cell in both specifications. For DROUGHT the rows show the all-drought (threshold-only) SPEI-12 coefficient — null — alongside the ENSO-aligned matched-gate coefficient, which is negative; they are shown per a $+10$ percentage-point change in the three-year cumulative exposure share, so the absolute scale is not directly comparable to the acute rows. The non-corroborating SSI-12 streamflow index is reported as appendix triangulation rather than in the forest plot.

6 Limitations

Single-episode identification for DROUGHT. The DROUGHT effect is identified from the post-2020 late megadrought: restricting the panel to pre-2020 output years turns the matched ENSO-aligned coefficient positive ($+0.80$), and only the post-2020 block carries the negative sign (-0.62). Within the sample the effect is robust to the matched-window teleconnection ρ_k, the gate definition, the fixed-effects ladder, and a lead placebo, and the all-drought-versus-ENSO-aligned decomposition holds; but a clean multi-episode identification would require a second slow climate event with different timing. The streamflow cross-index (SSI-12) does not corroborate the meteorological result and is reported as appendix triangulation. The channel is therefore read as ENSO-aligned persistent megadrought stress, not as a sample-wide drought law or as proof of ENSO causation of the megadrought.

Spec A vs Spec B is associational, not a causal instrument for ENSO. The contrast between General (Spec A) and ENSO-attributable (Spec B) tells us the sign and order of magnitude of the ENSO-aligned subset of physical flood/heat shocks, but is not a clean instrument for the ENSO channel itself: Spec B's comparison group pools non-ENSO threshold months with genuinely quiet months, which biases the coefficient toward zero by construction. At the mine-cell scale, the ENSO firing count is so low (1.5% of active mine-months, 65 firings after singleton filtering) that small-cluster inference cannot reject zero with seven regional clusters.

No mine-buffered corridor in the present construction. The current report uses the regional-mean monthly accumulation (headline) and the point-level mine cell (conservative lower bound); the natural intermediate spatial scale — a mine-buffered corridor defined around each mine's slurry pipeline, access road, and basin — is a substantial and promising idea but requires dedicated geospatial-attribution work (mine-buffered cell masks, distance-decay weighting, or basin-restricted polygons). It is reserved for a future revision.

HEAT methodology. The HEAT channel is constructed at monthly resolution (monthly-mean Tmax $\geq 23.5\,^{\circ}$C, calibrated to the daily DS 594/1999 27 $^{\circ}$C rule) paired with the monthly-mean Z. The mine-cell coefficient is a robust null. A finer-frequency operational outcome (worker absenteeism, shift productivity) would be needed to refine the HEAT specification further; monthly-frequency mine output cannot distinguish slow-integrator from daily-extreme HEAT mechanisms.

7 Reproducibility

1. **Climate ingest (scripts `10`, `11`, `12`).** `10` splices CR2MET 1960–2021 with the 2022–2025 extension at each mine cell and computes daily Z against the (mine, day-of-year) 1980–2010 baseline. `11` assembles the region-month ENSO context (ONI, regional teleconnection ρ, monthly Z). `12` aggregates daily to monthly mine-cell indicators: for FLOOD the threshold flag and the max daily Z on the wet day; for HEAT the threshold flag (monthly-mean Tmax $\geq 23.5\,^{\circ}$C) paired with the monthly-mean Z that is used in the ENSO-attribution step. The max daily Z on hot days is retained as an intermediate column but is not the HEAT Z used downstream.

2. **Master panel (script `20`).** Joins all primitives into the auditable mine $\times$ date $\times$ channel master panel ($19 \times 792 \times 5 = 75{,}240$ rows), written to `ledger/20__master_stress_ledger.csv` (CSV for inspection) and `.parquet` (for analysis).

3. **Regression panels (scripts `30`, `31`).** `30` builds the mine-month acute panel; `31` builds the mine-year DROUGHT panel.

4. **Regressions and inference (scripts `40`, `41`).** `40` runs the acute regressions with CR0 + WCB inference and the LOO-region / LOO-event battery; `41` runs the DROUGHT annual regressions.

5. **Validation (scripts `42`, `flood_validation/90`, `flood_validation/91`).** Classifies the 20 documented flood-disruption episodes into ENSO-related and Non-ENSO cohorts using NOAA ONI + regional teleconnection (independent of the AND-gate), then grades Spec A and Spec B as a binary

classifier (sensitivity / specificity).

6. **ENSO-attribution robustness (script 43).** Re-fits headline regressions under three alternative ENSO-attribution rules (seasonal-window ONI; Spearman ρ; phase-conditional $\overline{Z}$).
7. **Figures (scripts 60, 61).** `60` renders the map and the FLOOD validation chart; `61` renders the case-study chart, drought-profile chart, and coefficient forest plot.
8. **Tables (script 70).** Compiles LaTeX table fragments from the regression CSVs into `tables/`.

Every stress flag in the report can be re-derived by inspection from the master panel.

8 Conclusion

This report asked when an ENSO-related climate anomaly that crosses a documented operational threshold lowers physical copper output in the affected Chilean region. The headline finding is that ENSO risk is channel-specific rather than uniform: episodic for floods, absorbed at monthly resolution for heat, and cumulative for drought. No single coefficient summarises the climate–output relationship; each channel has its own time-scale, spatial geography, and inference profile.

For FLOOD, the headline regional-mean Spec A coefficient is -4.7% ($p_{\text{WCB}} = 0.013$): a wet regional month reduces monthly mine output by roughly $4-5\%$. The ENSO-attributable subset (Spec B) accounts for about half that effect at -2.5%, directionally consistent but not statistically resolved on the 7-cluster sample. The mine-cell variant, a conservative point-level lower bound, returns -1.8% (Spec A) and approximately zero (Spec B with singleton-firing mines filtered, since the AND-gate fires only 65 mine-months at the point level). The case-study annex confirms the physical mechanism at named events: large reductions (-15% to -35%) at cordillera mines during the June and August 2023 atmospheric-river / central-Chile flood events, with the regional-flood signature smaller but consistently directional across the panel. A mine-buffered corridor at an intermediate spatial scale is the natural next refinement; constructing it requires geospatial-attribution work reserved for a future revision.

For HEAT, the mine-cell coefficient is a robust null across all four ENSO-attribution rules. The firing pattern is bimodal at the extremes: four operations in the hot Antofagasta basin cross the monthly-mean threshold in essentially every month while the ten high-cordillera majors never do, with a handful of mid-altitude cases (Zaldivar, Candelaria, El Tesoro) crossing intermittently. The interpretation is constrained by monthly resolution: routine heat exposure at mines that face it constantly leaves no measurable footprint in monthly output, consistent with operations adapted to that exposure, but the data cannot speak to finer-frequency productivity losses (worker absenteeism, shift slowdown) that a monthly aggregate would average through.

For DROUGHT, the slow channel is read as ENSO-aligned persistent megadrought stress. The three-year cumulative share of *all* SPEI-12 drought months has no association with annual output ($\hat{\beta} = +0.075$); the output penalty is concentrated in the subset aligned with a matched ENSO-aligned teleconnection gate ($\hat{\beta} = -0.277$, $p = 0.025$ under mine + year FE; -0.529 with mine-specific trends; though randomisation inference places the honest small-sample p near 0.13), with the non-ENSO drought component null in the two-regressor decomposition. The result survives the matched-window teleconnection, the gate definition, the fixed-effects ladder, and a lead placebo, and is negative across the multi-month SPEI accumulation scales (statistically resolved at SPEI-3 and SPEI-12), consistent with a slow water-stress channel rather than a short-window artifact. It is, however, identified from the single post-2020 megadrought block (the pre-2020 sample is null), and the streamflow index SSI-12 does not corroborate it. The DROUGHT finding should be read as a real, time-scale-validated, ENSO-aligned persistent-drought effect within the megadrought, not as a general law that drought lowers mine output.

Two methodological points are worth stating in summary. First, the substantive ENSO content lives in the Spec A vs Spec B contrast, not in either coefficient in isolation. The General flood shock (Spec A) captures the average effect of a wet regional month, regardless of cause; the ENSO-attributable subset

(Spec B) isolates the share that passes the four-condition AND-gate. The contrast tells us *how much* of the physical flood effect is concentrated in ENSO-aligned events. Second, the AND-gate is conservative by design: the comparison group in Spec B pools non-ENSO threshold months with genuinely quiet months, which biases the estimated ENSO-attributable coefficient toward zero. Reported magnitudes should be read as lower bounds on the channel's true ENSO-attributable share rather than as point estimates of it.

Three avenues would tighten future work. A mine-buffered corridor for FLOOD — defined by a buffer around each mine's slurry pipeline, access road, and basin (rather than the broader regional mean or the narrower mine cell used here) — would lie at the intermediate spatial scale and is the natural next refinement; constructing it requires dedicated geospatial-attribution work. A finer-frequency operational outcome for HEAT (shift-level productivity, absenteeism) would let the monthly null be tested against the daily-extreme mechanism that monthly resolution cannot see. And a longer historical sample that contains a second multi-year water-stress episode independent of the 2010–2022 megadrought would convert the single-megadrought DROUGHT evidence into a genuine multi-episode identification. Each of these extensions targets a specific limitation called out above rather than reopening the headline result.

Annex A FLOOD case-study validation

The FLOOD stress signal is validated in two complementary steps. §A.1 tests whether the ENSO-attribution rule fires on documented Chilean flood-disruption episodes — the upstream test of the rule's credibility, independent of any mine-level outcome. §A.2 then asks whether the mines on which the rule fires actually respond at monthly resolution — the downstream reality check that explains the regression's panel-mean dilution.

A.1 Does the rule fire on ENSO-related flood events, and correctly reject non-ENSO floods?

The validation is set up as a binary-classifier test on 20 documented Chilean flood-disruption episodes 2015–2024. The episodes are drawn from the engineering and climate literature: the March 2015 Atacama aluvión (Wilcox et al. 2016 GRL), the April–May 2017 Cajón del Maipo cluster, the February 2023 Atacama / Coquimbo storm, the June 2023 atmospheric river, the August 2023 central Chile floods, and the April 2024 Maipo cluster. These events are selected for their physical magnitude — they caused mining-relevant disruption — not for their ENSO classification, so the validation needs an *independent* meteorological ground truth before grading the rule.

Ground-truth cohort assignment. For each event we classify the upstream meteorology in the event year-month independently of any threshold or alignment flag. An event is *ENSO-related FLOOD* when the NOAA ONI is in an officially active phase ($|\text{ONI}| \geq 0.5$) *and* the regional teleconnection times the ONI predicts wetter conditions ($\rho^{\text{pr}}_{r,m} \cdot \text{ONI}_{y,m} > 0$). An event is *Non-ENSO FLOOD* when those conditions fail — either the ONI is in the Neutral band or the teleconnection points the wrong way. An ONI in the precursor band $0.4 \leq |\text{ONI}| < 0.5$ with the teleconnection on-side is treated as *Borderline* and reported but not folded into either summary rate. Of the 20 documented events, this gives 13 ENSO-related, 6 Non-ENSO, and 1 Borderline per channel.

What we want to see. Spec A (any-cause threshold) should fire on most events in both cohorts — it is a physical-magnitude detector that does not know about ENSO. Spec B (the AND-gate stress flag) should fire on the ENSO cohort and stay silent on the Non-ENSO cohort; this is the classifier we actually grade.

Table 8: The 20 documented Chilean flood-disruption episodes 2015–2024 with NOAA ONI at the event month, the regional teleconnection's predicted direction, the ground-truth ENSO cohort, and the AND-gate's per-channel verdict. Cell codes: **S** = Spec B (ENSO-attributable) stress AND-gate fired (correct catch for ENSO; false positive for Non-ENSO); T = only the Spec A (general) physical threshold fired (correctly catches the flood as a physical event but does not classify it as ENSO); - = neither fired. Cohort cells: ENSO = $|\text{ONI}| \geq 0.5$ and $\rho \cdot \text{ONI} > 0$ (teleconnection predicts wet); Border = ONI in 0.4–0.5 precursor band; Non-ENSO = otherwise.

Event	Date	Reg.	ONI	Phase	Cohort	Mine-cell	Reg-mean
Mar 2015 Atacama aluvión	Mar 2015	ATA	+0.7	El Niño	ENSO	**S**	**S**
Mar 2015 Coquimbo spillover	Mar 2015	COQ	+0.7	El Niño	ENSO	-	**S**
Apr 2015 Atacama spillover	Apr 2015	ATA	+0.8	El Niño	ENSO	-	-
Aug 2015 central Chile rainfall	Aug 2015	RM	+1.9	El Niño	ENSO	**S**	**S**
Feb 2017 Atacama micro-storm	Feb 2017	ATA	+0.0	Neutral	Non-ENSO	-	-
Apr 2017 Coquimbo cluster	Apr 2017	COQ	+0.3	Neutral	Non-ENSO	-	-
Apr 2017 O'Higgins cluster	Apr 2017	OHI	+0.3	Neutral	Non-ENSO	T	T
Apr 2017 Cajón del Maipo I	Apr 2017	RM	+0.3	Neutral	Non-ENSO	-	T
May 2017 Cajón del Maipo II	May 2017	RM	+0.4	Neutral	Border	T	T
Feb 2023 Atacama / Coquimbo (Atacama)	Feb 2023	ATA	-0.3	Neutral	Non-ENSO	-	-
Feb 2023 Atacama / Coquimbo	Feb 2023	COQ	-0.3	Neutral	Non-ENSO	-	-
Jun 2023 atmospheric river Valparaíso	Jun 2023	VAL	+0.8	El Niño	ENSO	**S**	T
Jun 2023 atmospheric river Maule	Jun 2023	OHI	+0.8	El Niño	ENSO	**S**	T
Jun 2023 atmospheric river O'Higgins	Jun 2023	OHI	+0.8	El Niño	ENSO	**S**	T
Jun 2023 atmospheric river RM	Jun 2023	RM	+0.8	El Niño	ENSO	**S**	T
Aug 2023 floods Valparaíso	Aug 2023	VAL	+1.4	El Niño	ENSO	**S**	T
Aug 2023 floods Maule	Aug 2023	OHI	+1.4	El Niño	ENSO	**S**	**S**
Aug 2023 floods O'Higgins	Aug 2023	OHI	+1.4	El Niño	ENSO	**S**	**S**
Aug 2023 floods RM	Aug 2023	RM	+1.4	El Niño	ENSO	**S**	T
Apr 2024 Maipo cluster	Apr 2024	RM	+0.8	El Niño	ENSO	-	T

Table 9: Sensitivity and specificity of the FLOOD detectors against the ground-truth cohorts in Table 8. Sensitivity = share of ENSO-related events caught; specificity = share of Non-ENSO events correctly rejected. Spec A = physical threshold only; Spec B = AND-gate stress flag.

Channel	**n** (ENSO / Non-ENSO)	**Spec A (threshold)** Sens.	Spec.	**Spec B (stress AND-gate)** Sens.	Spec.
FLOOD reg-mean monthly (headline)	13 / 6	92.3%	66.7%	38.5%	100.0%
FLOOD mine-cell (conservative)	13 / 6	76.9%	83.3%	76.9%	100.0%

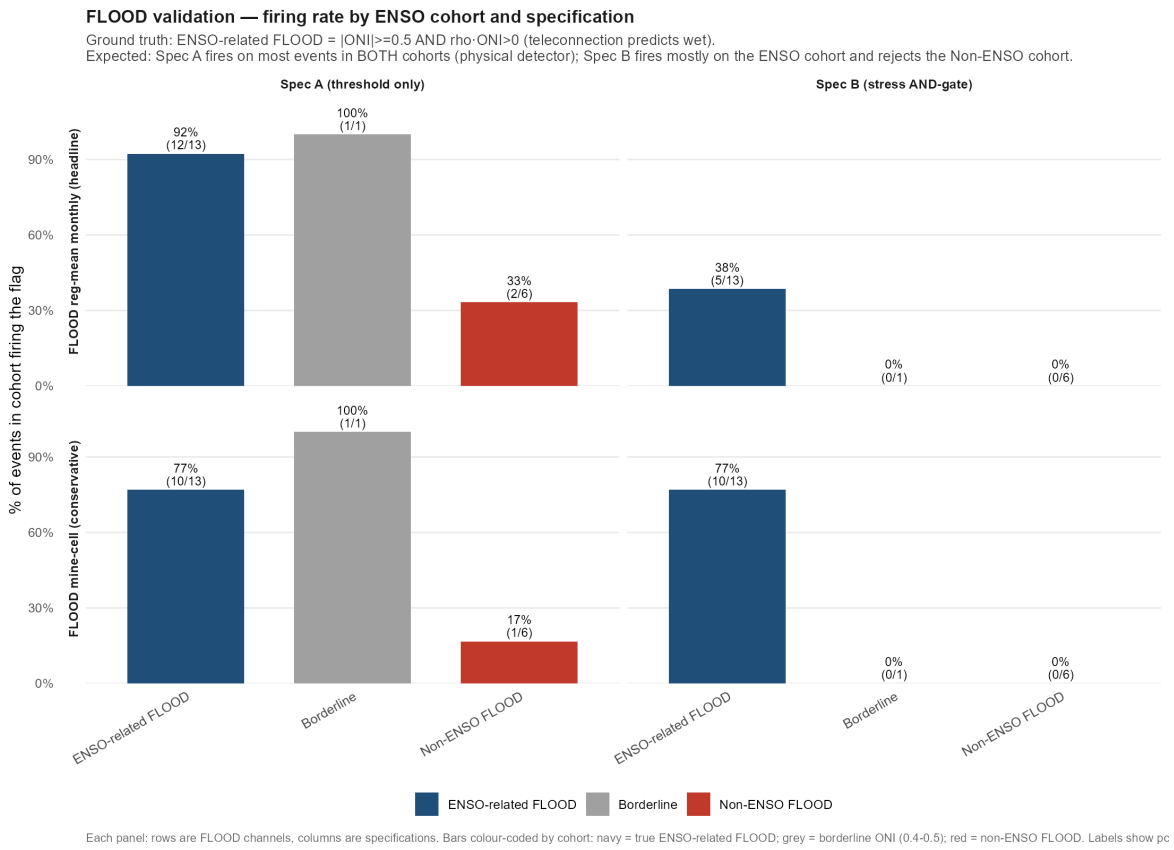


Figure 4: FLOOD validation re-framed as a binary classifier test. Rows: the two retained FLOOD variants. Columns: Spec A (threshold only) and Spec B (AND-gate stress flag). Bars colour-coded by ground-truth cohort: navy = ENSO-related FLOOD; grey = Borderline ONI; red = Non-ENSO FLOOD. Labels are firing percentage (count of firings / events in the cohort). The cleanest reading: **Spec B fires on the ENSO cohort and rejects the Non-ENSO cohort at every channel** (specificity = 100% on every channel); **Spec A fires across both cohorts**, confirming it is a physical-magnitude detector.

Headline numbers from Table 9. At the FLOOD mine-cell variant (the conservative point-level lower bound), the AND-gate's sensitivity is 76.9% (10 of 13 ENSO-related events caught) and its specificity is 100% (0 of 6 Non-ENSO floods falsely flagged). At the FLOOD regional-mean monthly variant (the report's headline), the AND-gate's sensitivity is 38.5% — much lower — though specificity is also 100%. The lower sensitivity at the regional mean reflects the resolution-mismatch problem: a 3-day storm in the middle of a normally-wet austral-winter month averages out to a monthly-mean Z that fails the sign condition of the AND-gate. The mine-cell variant is therefore the stronger *ENSO classifier*, but its low firing count limits its identifying power in the regression; the regional-mean variant is the better *physical-shock detector* for the regression at the cost of weaker ENSO classification at the AND-gate.

The Non-ENSO column of Table 8 shows what Spec A is doing: it catches some of the Non-ENSO floods (2/6 at the regional mean; 1/6 at the mine cell) and correctly rejects others (e.g., the February 2023 events, both Atacama and Coquimbo, did not in fact reach the operational threshold in the precipitation data). This is the right division of labour: Spec A grades physical magnitude (general flood shocks), Spec B grades ENSO causation (ENSO-attributable subset), and the two specifications combined identify both *whether* a flood happened and *whether* it was teleconnected to an active ENSO phase.

This first validation is upstream of the regression: it tests whether the rule classifies the right events, not yet whether those events leave a measurable footprint in monthly mine output. The downstream question is addressed next.

A.2 When the rule fires, do mines actually respond?

The case-study chart (Figure 5) reports percentage deviations from each mine's 2006–2025 calendar-month baseline at five documented ENSO-related flood episodes 2006–2025. Several deviations are positive — in particular, the April–May 2017 central-Chile cordillera mines operated at $+8\%$ to $+16\%$ of their long-run baselines during widely-documented lowland floods. Four mechanisms explain why those positive deviations are informative rather than contradictory, and the aggregate reading validates the headline FLOOD identification.

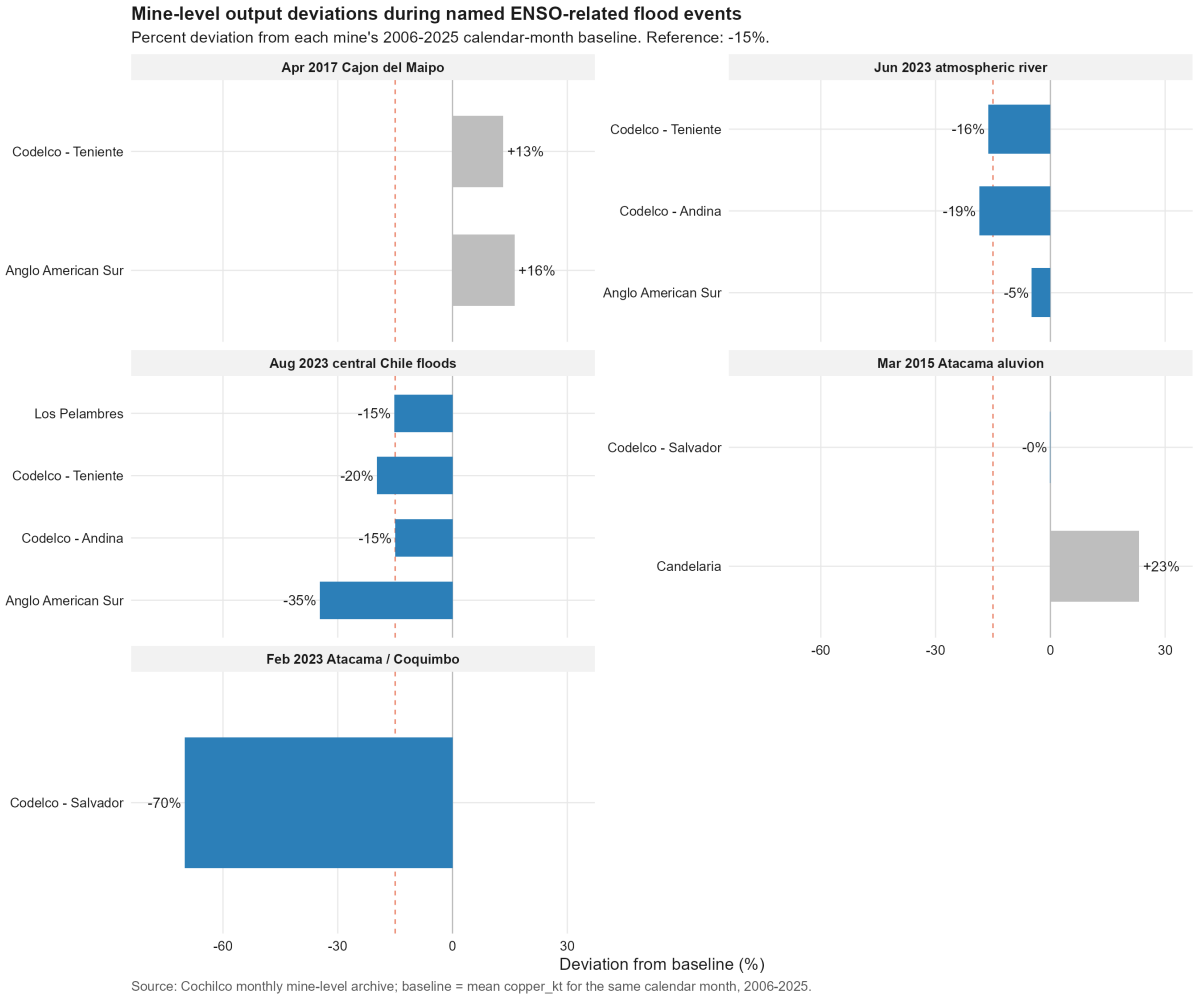


Figure 5: Mine-level output deviations during named ENSO-related flood events 2006–2025. Percentage deviation from each mine's own 2006–2025 calendar-month baseline; dotted reference line at -15%. Positive deviations during the 2017 lowland cluster reflect the precipitation not reaching the high-altitude mine sites; the largest negative deviations are at the high-cordillera mines during the 2023 atmospheric-river and central-Chile flood events.

(1) The flood did not reach the mine site. The April 2017 Cajón del Maipo event devastated populated lowland Santiago — valley roads, water infrastructure, lower-Maipo agricultural land. But Codelco-Teniente ($\approx 2{,}800$ m), Codelco-Andina ($\approx 3{,}500$ m), and Anglo American Sur ($\approx 3{,}500$ m) sit at altitude; at the mine sites the precipitation fell as snow or did not exceed any operational threshold, and none reported operational disruption in the Cochilco archive. This is the substantive finding the main report rests on for FLOOD: the panel-average coefficient is small because documented regional floods often do not bind at altitude. Contrast with the 2023 events, which *did* deposit precipitation at altitude and produced the -15% to -35% reductions in the bottom row of Figure 5.

(2) Year-trend / capacity-ramp baseline effects. The 20-year baseline averages across a span over

which Codelco-Andina (mid-2010s plant-debottlenecking) and Anglo American Sur (post-2010 expansion) were in measurable growth phases. The headline regression absorbs these via year fixed effects; the chart's simple multi-year baseline does not. So the 2017 positive deviations partly reflect mine-specific growth overlaid on the (real) fact that the floods did not hit the high-altitude cells. The case-study chart is *complementary to* rather than *a substitute for* the regression.

(3) Sub-monthly disruption averaged into a full-month figure. The March 2015 Atacama aluvión struck on March 24–26. Candelaria's March output was $+23\%$ above its long-run March mean: the disruption was confined to the last week, the mine recovered before month-end, and the monthly aggregate averaged through it. Codelco-Salvador, in the same Atacama region, shows essentially zero deviation (-0.1%) for the same reason — the aluvión disrupted Copiapó-valley logistics but the mine itself kept producing. The case shows that monthly resolution is too coarse to capture sub-monthly operational disruption; the limitations section flags this explicitly.

(4) Coincident operational transitions. A separate mechanism applies to Codelco-Salvador's -70% in February 2023. The headline number reads as a dramatic flood response, but Salvador was already at the bottom of its production cycle in early 2023 — the prelude to the Rajo Inca modernisation that took the mine to zero output for January–September 2024. The pre-modernisation production decline overlaps with the February 2023 Atacama / Coquimbo storm, so the -70% partly captures the operational transition rather than a pure flood effect. The case is informative about a generic limitation of any month-frequency comparison: large isolated negative deviations need to be screened against the mine's contemporaneous operational state before being attributed to a climate event.

Aggregate reading and methodological implication. The largest mine-level reductions from *clean* ENSO-related flood events occur at August 2023 (Anglo American Sur -35%, Codelco-Teniente -20%, Codelco-Andina -15%) and June 2023 (Codelco-Andina -19%, Codelco-Teniente -16%). These are large, operationally meaningful, named-mine effects of named ENSO-related events with no confounding operational transition. Together with §A.1, the two-step validation tells one coherent story: the ENSO-attribution rule (§2.4) identifies *which* flood-threshold violations are ENSO-related, and the case studies confirm that the rule's identified events include the operationally most consequential disruptions (2023 cordillera) without over-attributing to operationally null lowland-only events (2017 cluster), while flagging Codelco-Salvador February 2023 as a case where the mine-level deviation reflects coincident operational decline rather than pure flood impact.

Annex B Hydrological SSI-12 cross-check (not retained in the main specification)

A hydrological streamflow drought index, SSI-12, was explored as a cross-check on the slow water-stress channel but is not retained in the main specification. SSI-12 measures source-basin streamflow rather than mine-cell meteorological water balance, and representative gauges are unavailable or physically inappropriate for many operations: closed endorheic salars with no representative streamflow gauge (Escondida, Zaldívar, Gabriela Mistral, El Abra), operations on desalinated or direct-seawater supply decoupled from any continental basin (Candelaria, Quebrada Blanca, Esperanza, El Tesoro), and the coastal cross-region Anglo American Norte aggregate. On the small gauged-basin subset it does not corroborate the SPEI-12 result: under the same matched ENSO-alignment gate the streamflow coefficient is positive rather than negative ($\hat{\beta} = +0.501$, $p = 0.065$, $N = 148$). Because SSI measures a physically different exposure, this is treated as a data limitation and triangulation note rather than as required corroboration of the meteorological channel.